Asset Performance:

Ene 15 Monitoring energy end uses

4 Credits

0

☆☆☆☆☆☆
☆☆☆☆☆
☆☆☆☆
☆☆☆
☆☆
☆

No Minimum Standard

Aim

To reduce operational energy consumption through the effective management and monitoring of energy consumption for different building services and building services systems.

Question

What percentage of end uses with significant energy consumption are sub-metered?

Credits	Answer	Select a single answer option
0	A.	Question not answered
-	B.	There are no end uses with significant energy use
4	C.	100% of end uses with significant energy consumption are sub-metered
3	D.	≥75% of end uses with significant energy consumption are sub-metered
2	E.	≥50% of end uses with significant energy consumption are sub-metered
1	F.	≥25% of end uses with significant energy consumption are sub-metered
0	G.	<25% of end uses with significant energy consumption are sub-metered

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	**Filtering** Where there are no end uses with significant energy consumption in the asset the associated credits can be filtered out of the assessment.	B
2.	Significant energy consumption is considered to be >8,500kWh per year for electricity and >67,000kWh per year for other energy sources	C - G
3.	Where there is more than one HVAC system type serving a particular end use the requirement applies to the main HVAC system.	C - G

Criterion	Assessment criteria	Applicable Answer
4.	Where more than one end use is provided by the same HVAC component it is acceptable to sub-meter the combined energy consumption.	C - G

Methodology

Determining significant energy consumption

1. Identify which of the following end uses are present in the asset;
   a. Space heating generation
   b. Space cooling generation
   c. Hot water generation
   d. Mechanical ventilation
   e. Fans for distributing space heating
   f. Fans for distributing space cooling
   g. Pumps for space heating
   h. Pumps for cooling
   i. Pumps for hot water
   j. Commercial scale refrigeration
   k. Internal lighting
   l. Controls and telecommunications
   m. IT equipment and small plug in loads
   n. Internal transport (lifts and escalators)
   o. External lighting
   p. Other (user defined)

2. For each fuel type used in the asset estimate the energy consumption to determine whether it is significant;
   a. End use energy consumption may be estimated based on installed capacity and anticipated full load run hours, based on measured energy consumption data for the assessed asset for a similar asset.
   b. Where estimates of energy consumption are not available the BREEAM In-Use Online platform will provide an indication of end uses that are significant based on default values.
   c. In instances where more than one end use is provided by a single building servicing system end use energy consumption may be estimated based on installed capacity and anticipated full load run hours, based on measured energy consumption data for the assessed asset or for a similar asset.
   d. The BREEAM In-Use default end use energy consumption values are calculated based on the floor area of the asset and pessimistic energy consumption values which are higher than the typical values for the end use.

Calculating energy use by subtraction

It is acceptable to calculate energy consumption for an end-use by subtracting sub-metered energy consumption for other end-uses from the relevant main utility meter reading.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Estimated energy consumption for building servicing systems indicating the end uses that they provide
All	Either copies of verified sub-meter data for the first and last date of the 12-month period specified. These may be outputs from energy monitoring and management systems or automatic or manual meter readings OR; Line diagram indicating sub-meters and the related energy uses or evidence to show that the end uses can be monitored separately.

Definitions

Small plug in loads:

Plug-in equipment/appliances connected through power points.

Energy monitoring and management system:

Examples include automatic meter reading systems and building energy management systems (BEMS). Automatic monitoring and targeting is an example of a management tool that includes automatic meter reading and data management.

Significant energy consumption:

An end use or building servicing system is considered to be significant where the typical energy cost savings achieved through sub-metering is expected to payback within 10 years through energy cost savings achieved through improved energy management. For the purpose of this issue, energy use is where the estimated energy consumption exceeds the threshold kWh/year values of 8,500 kWh/year for electricity and 67,000 kWh/year for other fuels are deemed to be significant.

Sub-metering:

Sub-meters are secondary to the main utility meters and are installed to measure consumption by specific items of plant or equipment, or to discrete physical areas, e.g. individual buildings, floors in a multi-storey building, tenanted areas, functional areas. Outputs include pulsed outputs or other open protocol communication outputs.